

Building Detection from Satellite Imagery

Assignment 2 — YOLOv8 Object Detection

SpaceNet Paris Dataset | Vaibhavi Singh 251159

1. Problem

This project builds an automated building detection system for satellite imagery using deep learning. Detecting buildings from aerial images has real-world applications in urban planning, disaster response, population estimation, and infrastructure monitoring. Manual annotation of satellite images is slow and expensive, making an automated YOLOv8-based pipeline highly valuable for scaling these tasks.

2. Data

The dataset used is the SpaceNet 2 Paris Buildings dataset, sourced from Kaggle (ugorjiir/spacenet-2-paris-buildings). It contains RGB pan-sharpened satellite images (.tif) of Paris along with building footprint annotations in GeoJSON format. The raw data was converted to YOLO format: images were saved as .jpg files and building polygon annotations were converted to normalized bounding boxes in .txt label files.

Split	Images	Background Images
Train	918	414
Validation	115	55
Test	115	—

Each label file contains one line per building in the format: *class x_center y_center width height* (all normalized to 0–1). The dataset has a single class: building (class 0).

3. Model

YOLOv8 (You Only Look Once, version 8) is a state-of-the-art real-time object detection model developed by Ultralytics. It uses a single-pass convolutional neural network to predict bounding boxes and class probabilities simultaneously, making it significantly faster than two-stage detectors. The architecture includes a CSPDarknet backbone, a Feature Pyramid Network (FPN) neck for multi-scale detection, and a decoupled detection head.

We selected **YOLOv8s (small)** as a balance between speed and accuracy — large enough to learn complex building shapes, but small enough to train efficiently on a T4 GPU within the Colab

session limit.

Property	Value
Model variant	YOLOv8s
Total parameters	11,135,987
GFLOPs	28.6
Pretrained on	COCO dataset (80 classes)
Number of classes (finetuned)	1 (building)
Input image size	640 x 640

4. Training

The model was fine-tuned from pretrained COCO weights using the Ultralytics training pipeline. Transfer learning was used instead of training from scratch because the COCO pretrained weights already encode general visual features (edges, shapes, textures) that transfer well to satellite imagery.

Hyperparameter	Value
Epochs	30
Batch size	16
Image size	640
Optimizer	AdamW (auto-selected)
Learning rate	0.002
GPU	Tesla T4 (14 GB)
Training time	~0.164 hours (~10 minutes)
Patience (early stopping)	10 epochs

5. Results

The model was evaluated on the validation set after training. The final metrics from the best checkpoint (epoch 27) are shown below:

Metric	Value	What it means
mAP@50	0.818	Model detects buildings correctly 81.8% of the time at IoU=0.5
Precision	0.827	Of all predicted boxes, 82.7% were actual buildings
Recall	0.757	Of all real buildings, the model found 75.7% of them
F1 Score	0.790	Balanced score between precision and recall
MAE (count)	2.60	On average, predicted count differs from true count by 2.6 buildings

The training curves showed consistent improvement in mAP@50 from 0.616 at epoch 1 to 0.821 by epoch 27, with both train and validation losses decreasing steadily — indicating healthy learning without overfitting. The Precision-Recall curve showed a strong area under curve of 0.818, and the confusion matrix confirmed 1271 correct detections against 321 false positives and 308 false negatives.

6. Observations

What worked well:

Transfer learning from COCO weights gave the model a strong starting point, achieving mAP@50 above 0.60 even in the very first epoch. The model handled images with varying building densities well, with a mean absolute error of only 2.6 buildings per image on the validation set. Training converged quickly in under 10 minutes on a T4 GPU.

What the model struggled with:

The model had difficulty with densely packed buildings where rooftops overlap or touch — it tended to merge adjacent buildings into a single bounding box. Images with significant shadows from tall structures caused some buildings to be missed entirely. Very small rooftops (under ~10 pixels) were often skipped due to the 640px input resolution.

One idea to improve it:

Training with a larger model variant (YOLOv8m or YOLOv8l) combined with test-time augmentation (TTA) would likely push mAP above 0.85. Alternatively, tiling the large satellite images into smaller overlapping patches before inference would help detect smaller buildings that currently fall below the detection threshold at 640px resolution.